Physician Segmentation & Commercial Prioritization

A Pharma Commercial-Analytics Case Study

Executive Summary

A new heart-and-metabolism drug ("NovaCard") was being prescribed by only **10.2% of doctors**, even though the company was spending heavily to make it available. Using data on 800 doctors, I split them into three clear groups, used a simple machine-learning model to find out what really makes a doctor prescribe the drug, and turned that into a clear action plan. The main insight changed how to see the problem: the biggest group of doctors isn't skipping the drug because they don't know about it — it's because the drug is too costly or hard to get for their patients. So the smartest move is to focus on a smaller group that is ready to prescribe but isn't being reached, and fix the cost problem before spending more on the rest.

1. Business Context & Problem

Pharma sales teams have limited resources — sales-rep time, digital campaigns, and support to get the drug covered by insurance. The key question is simple: **which doctors should we focus on to grow the drug the fastest?** Trying to reach every doctor wastes money on people who will never prescribe, and misses the ones who are ready. The company needed a clear, data-backed way to group its doctors and decide where to spend next.

Objective: Group the doctors, find out what makes them prescribe the drug, and recommend where to focus limited resources to get the most prescriptions for the money spent.

2. Data

The analysis uses a **made-up (synthetic) dataset of 800 doctors**, built to behave like real pharma data. Real doctor-prescription data is hard to get and usually locked behind licences, so I created the data with realistic patterns built in — doctors prescribe more when reps visit them, when they are active online, and when their peers prescribe; and they prescribe less when the drug costs their patients more or when they are loyal to a rival drug. I keep this clear on purpose: it shows the full method without pretending to have private company data, and every column can be explained.

Key fields: specialty, region, number of patients, % of patients on Medicare (a cost-sensitivity signal), how easily the drug is covered by insurance (formulary tier), how often reps visit, online-engagement score, peer-influence score, competitor-loyalty score, patient out-of-pocket cost, and whether the doctor prescribed the drug.

3. Approach

- **Grouping the doctors —** I used K-Means clustering to group doctors by six behaviours (rep visits, online activity, peer influence, competitor loyalty, patient cost, and drug coverage). I then studied each group and gave it a plain-English name instead of leaving it as a number.
- **Finding what drives prescribing —** I trained two models — logistic regression and Random Forest — to tell prescribers apart from non-prescribers (accuracy scores of 0.81 and 0.75). Using two models is a cross-check: both point to the same top drivers, so the result isn't a fluke of one method.
- **Deciding where to focus —** I plotted the groups on a chart of market size vs. current prescribing, so the trade-off (a big untapped group vs. an easy-to-win group) is clear at a glance.

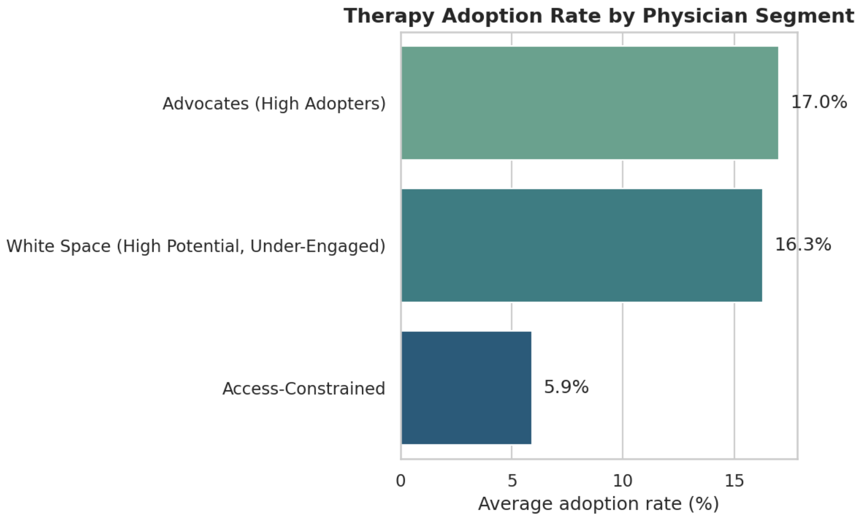

Figure 1. Prescription rate by doctor group. Advocates and White Space prescribe at about 3× the Access-Constrained rate.

4. Findings

Three clear, useful groups came out of the data:

Group	Doctors	Prescribing	Eligible Patients	What they look like
Advocates	176	17.0%	32,625	Reps visit often, drug is easy to cover; already prescribing.
White Space	203	16.3%	36,268	Easy coverage and strong peer influence, but reps rarely visit (~2 visits/quarter).
Access-Constrained	421	5.9%	77,399	Biggest group (~half the market); 44% have hard drug coverage and the highest patient cost ($103 avg).

What drives prescribing

Both models agree on the top drivers (the + or – shows the direction): **peer influence (+)**, **competitor loyalty (–)**, and **share of cost-sensitive / Medicare patients (–)**. Interestingly, how often a rep visits matters *less* than peer influence — what other doctors are doing counts for more than how often a rep shows up.

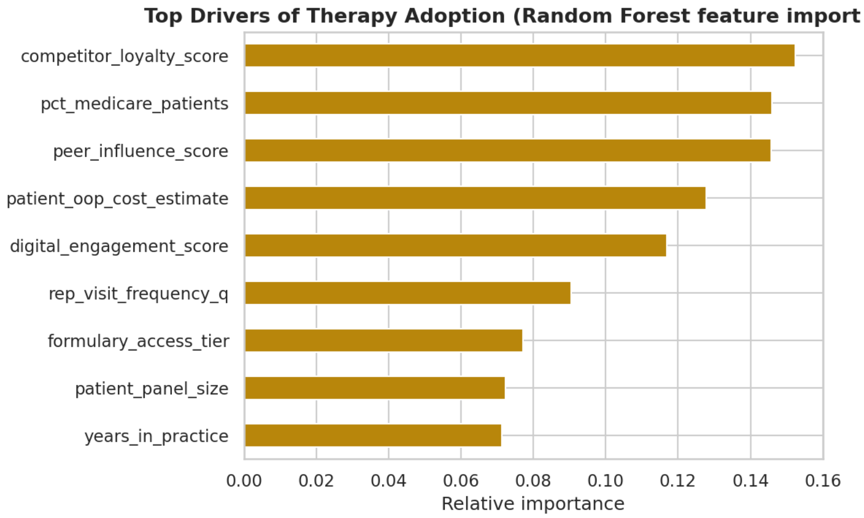

Figure 2. Which factors matter most (Random Forest). Peer, competitor, and cost/coverage factors lead.

The cost-and-coverage signal

Prescribing drops from **18.8%** when the drug is easy to cover (Preferred tier) to **8.9%** (Standard) to **4.2%** (Restricted). The Access-Constrained group is packed into these harder, costlier tiers. This is the proof that the biggest group's low prescribing is a cost-and-coverage problem, not a lack of awareness.

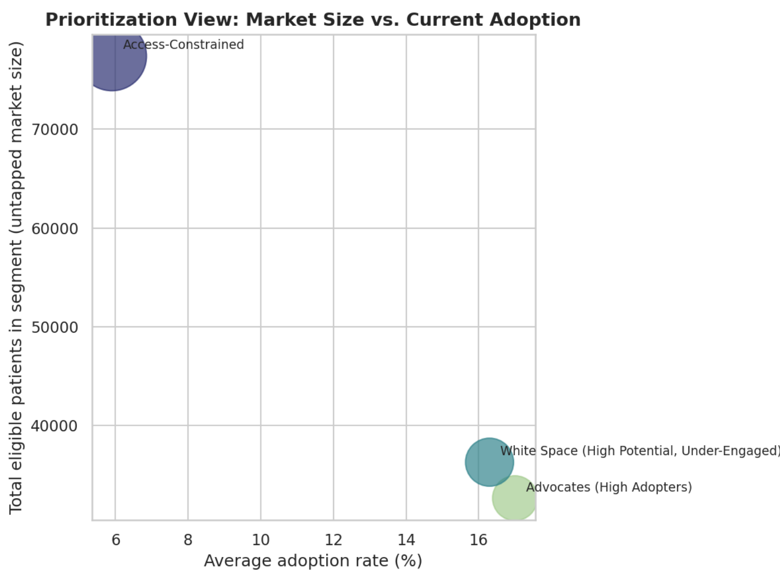

Figure 3. Where to focus — untapped patients vs. current prescribing; bubble size = number of doctors. Access-Constrained sits top-left: a huge untapped group with low prescribing.

5. Recommendation

1. Focus on White Space first. These 203 doctors already find the drug easy to prescribe and their peers prescribe it — but reps barely visit them. Visiting them more, plus peer/KOL programs, will win them over fastest and at the lowest cost. This is the best return for the money.

2. Fix Access-Constrained before spending more on it. This is the biggest group (about half the patients), but the problem is cost and coverage, not awareness. More rep visits won't help — it needs an access fix first, like co-pay support and help getting insurance approvals.

3. Keep Advocates happy, don’t over-spend. They already prescribe at 17% — keep them with light-touch effort instead of using up scarce rep time here.

4. Skip doctors loyal to a rival drug. Competitor loyalty is the single biggest reason a doctor won’t switch, so leave the most loyal ones out of the first push — they are unlikely to change even after the cost problem is fixed.

6. Business Impact

This plan moves limited resources away from spreading thin, and toward the best-return group (White Space) and the biggest group once it is unblocked (Access-Constrained). The key skill here — knowing when to ‘engage more’ versus ‘fix access first’ — is exactly what separates smart spending from wasted budget, and it is the same thinking ZS uses for pharma sales and patient-access problems.

7. Limitations & Next Steps

- The data is made-up. The next step is to test the grouping on real doctor data and check that the same drivers still hold.
- I chose the number of groups for clarity; a formal test (silhouette / elbow) would confirm it, and adding time (prescribing over quarters) would show urgency.
- Adding money — expected revenue per new patient — would turn the priority list into a clear dollar ranking.

8. Tools & AI-Assisted Workflow

Tools: Python (pandas, scikit-learn) for the data work, clustering, and models; Tableau Public for the interactive 3-page dashboard. I used AI coding tools to build faster, but I reviewed and own all the modelling choices and conclusions — a workflow where AI speeds up the work while the human stays in charge of the thinking.
